Mobile App Design Project

Brainstorming



2024 - 2025

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Structure of the course



Brainstorming Session

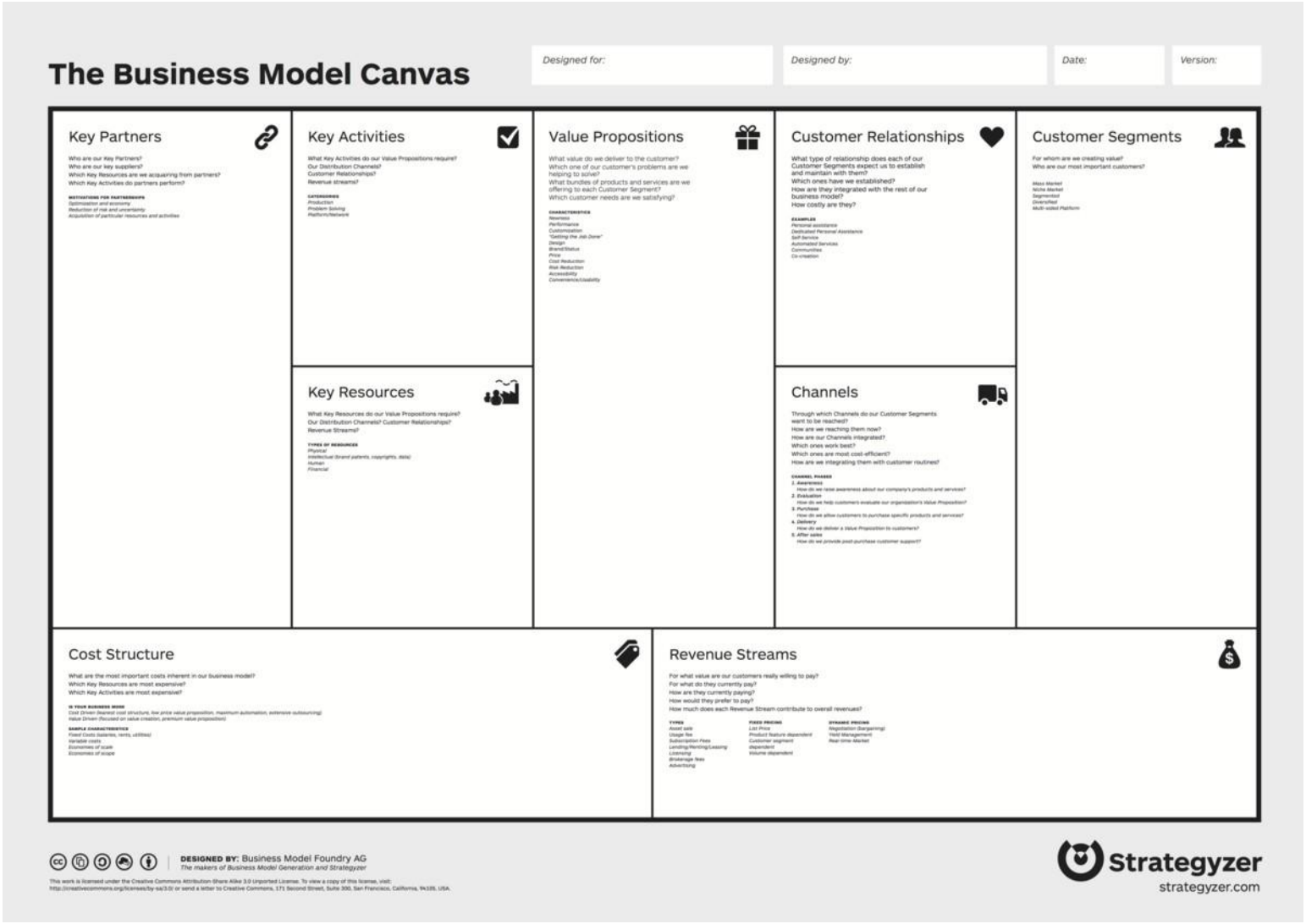
- Prerequisite: groups of 4-5 students via Webcampus formed

- Guidelines about:
  - How to structure your mobile app project using the Business Model Canvas ?
  - How to take an user-centered perspective ?
  - How to brainstorm for an innovative mobile app idea ?
  - Which tools to exploit in order to develop your project ?
  - How to pitch your idea?

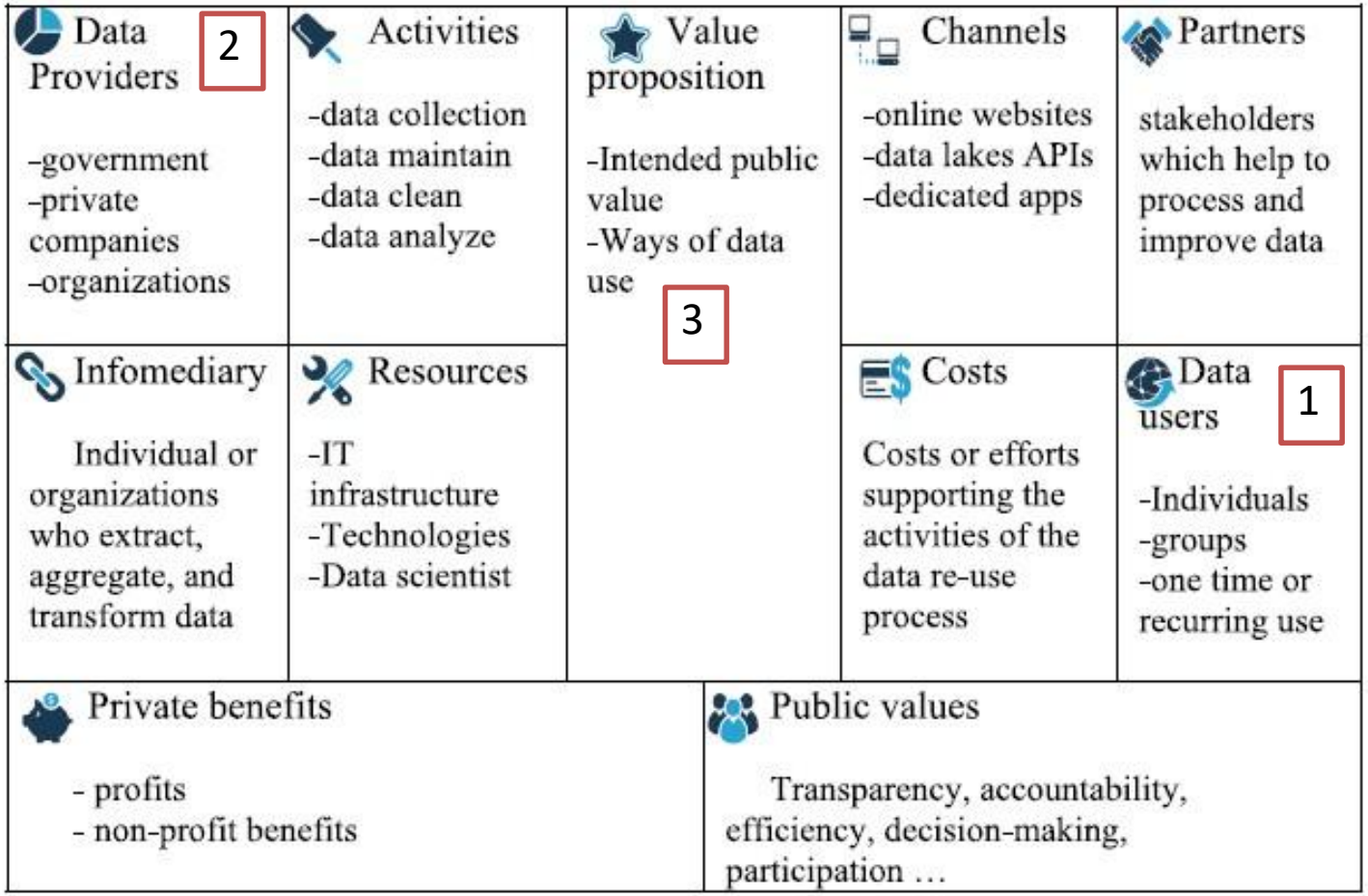
- You receive feedback and coaching about your preliminary ideas

STRUCTURING

Business Model Canvas … again



Where to start ? e.g. Open Data Canvas



1) Mobile app Users: Personas

- Ask yourself who are the users of the solution you are developing?
  - Who are they?
  - What are their roles?
  - What are their goals?

- Represent them with a persona

2) Applications Domain

CATALOGUE TRIÉ PAR THÈME

Thème

Aménagement du territoire, Urbanisme, Bâtiments, Equipements, Logement	49
Population, Statistiques	43
Administration, Gouvernement, Finances publiques, Citoyenneté	20
Transports, Déplacements	19
Culture, Patrimoine	13
Environnement	10
Santé	8
Sport, Loisirs	8
Energie	5
Economie, Business, PME, Développement économique, Emploi	4
Closed Data, Accès restreint	3
Education, Formation, Recherche, Enseignement	2

› Moins

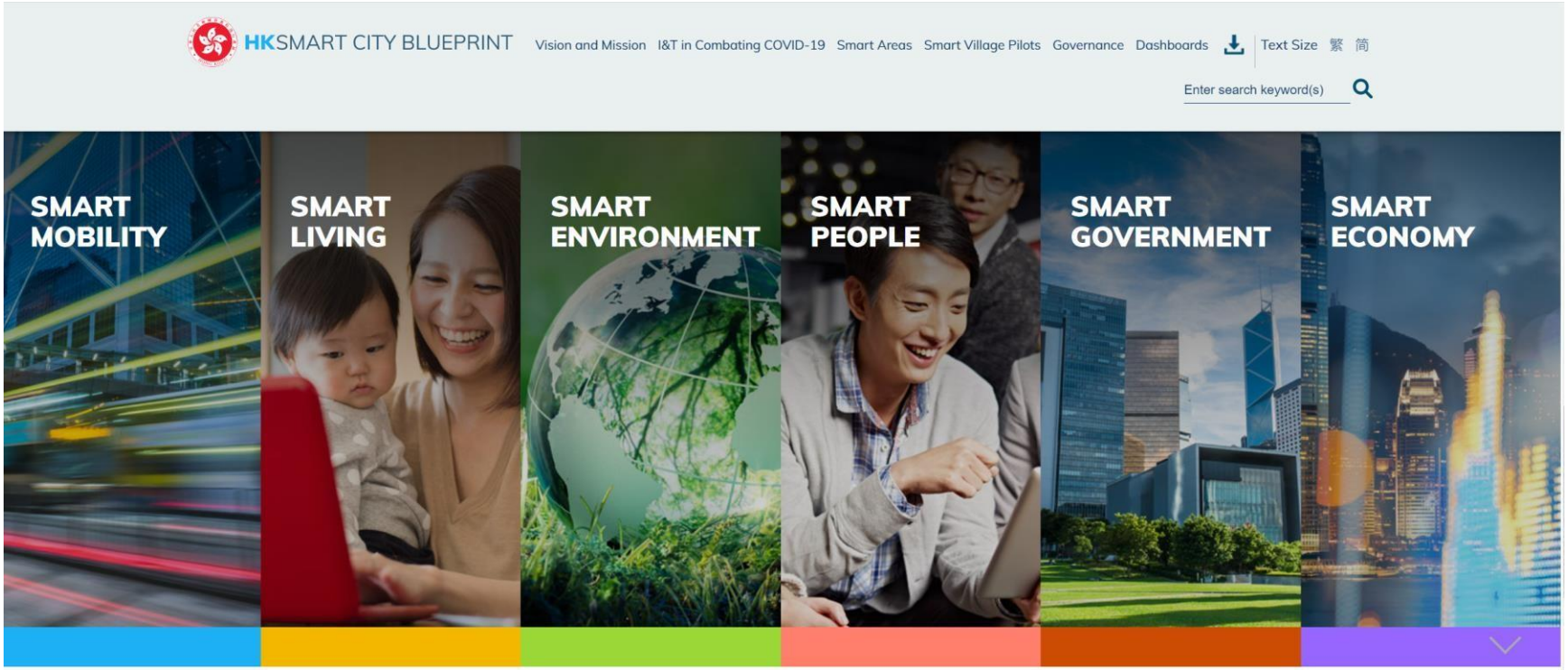
3) Value Propositions of the Mobile app







Example: Smart City

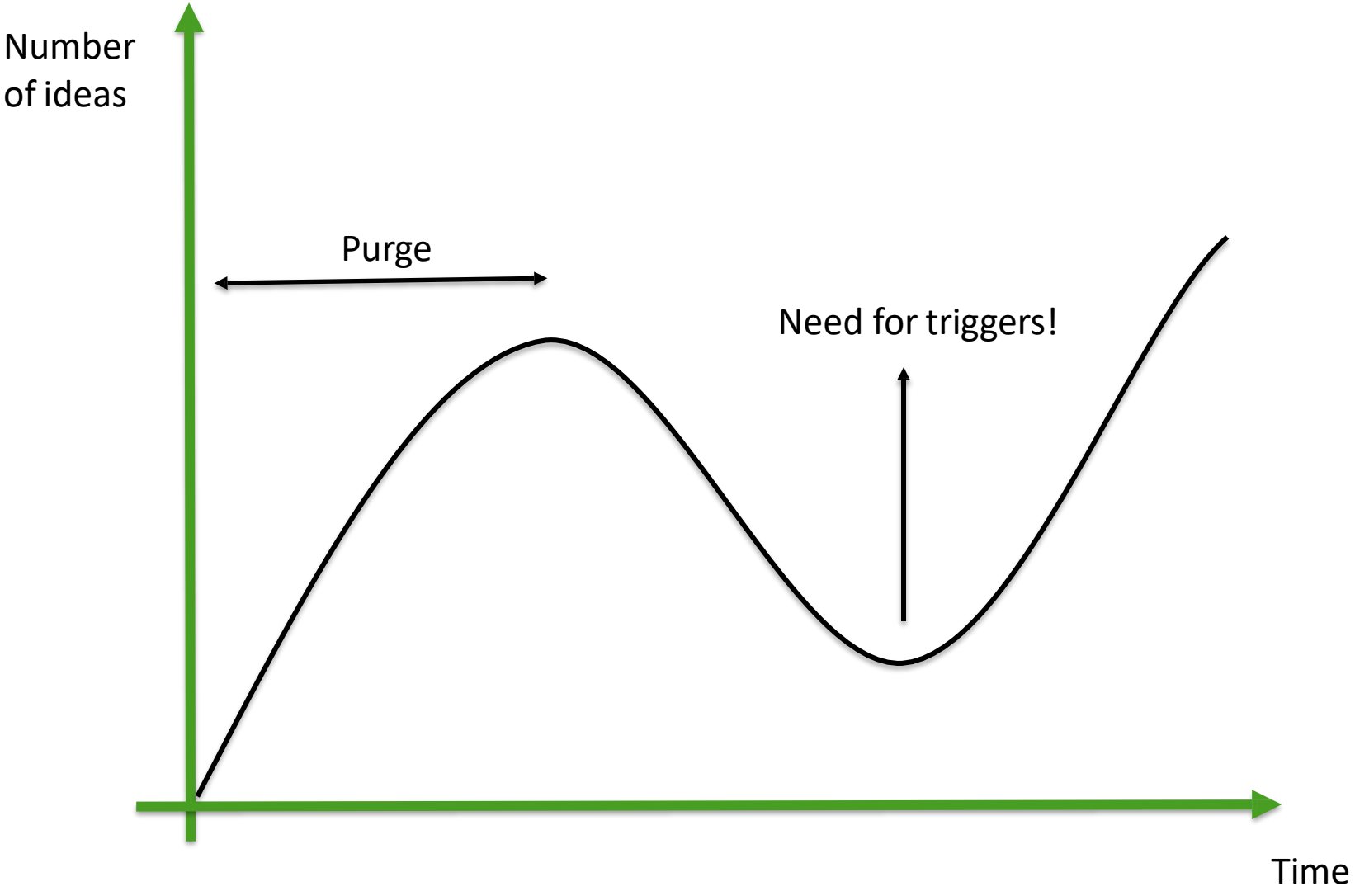


BRAINSTORMING

Brainstorming principles

- Go for quantity
- Withhold criticism (negative or positive)
- Welcome wild ideas
- Rebound, combine and improve ideas

Brainstorming takes time!



Trigger Cards and What-ifs

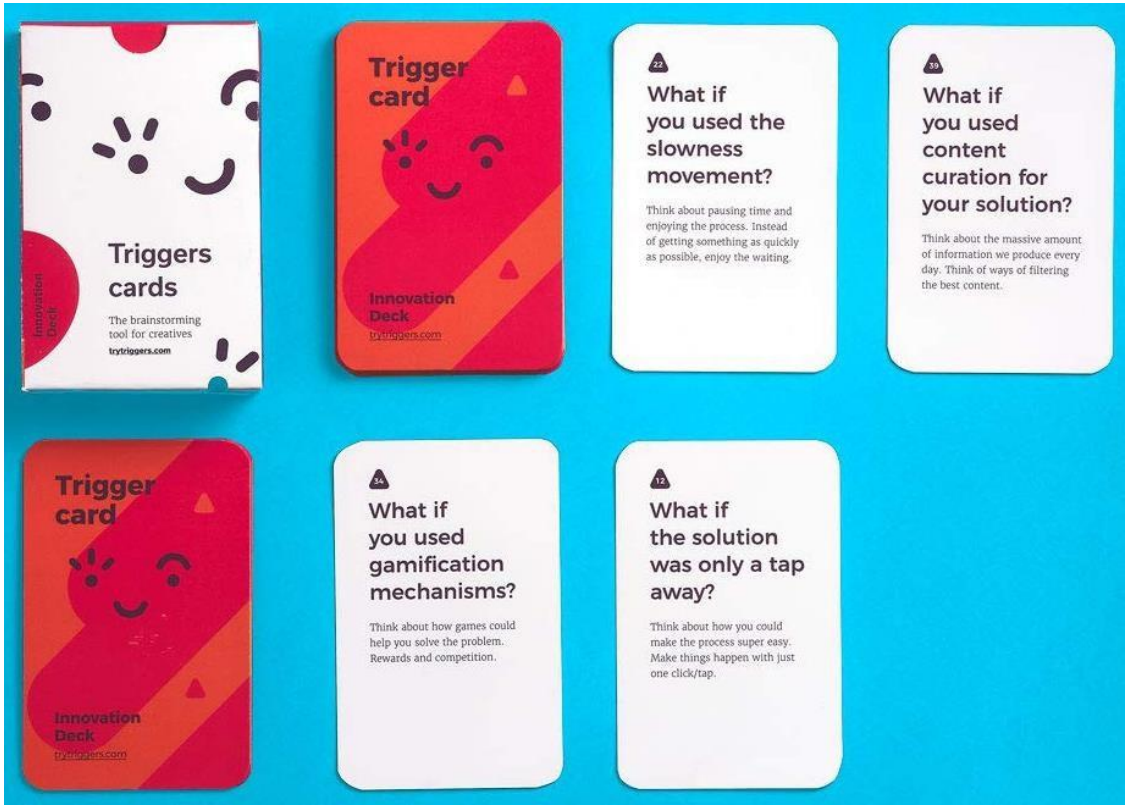
- Trigger Cards



- Online What-Ifs
  - https://whtifs.com/

Trigger cards

- **Innovation Deck** : focuses on new technologies and how they can improve your ideas



Trigger cards

- **Human-Centric Deck** : focuses on people and how your idea can add value for them



Six Hats Techniques



Blue Hat - Process

What conclusions or summaries can we make in moving forward on the issue or problem?



Yellow Hat - Benefits

What are the positive aspects of the current situation, issue or problem?



Green Hat - Creativity

What is the current information on issue or problem?



Black Hat - Cautions

What are the negative aspects of the current situation, issue or problem?



Red Hat - Feelings

How does everyone feel about the current situation, issue or problem?

White Hat - Facts

What is the current information on the issue or problem?

Suggested Process

1. Define problem (e.g. smart city dimension, user need)
2. Think possible solutions
3. Use a trigger (What-ifs, Persona, Hats, ...)
4. Refine/Brainstorm/Discuss

Choosing an idea: Mindmaps

- State the problem
- Brainstorm ideas
- Record ideas
- Sort ideas
- Identify common themes or categories
- Create the diagram based on ideas and categories
- Tool: https://klaxoon.com/fr/

Choosing idea: Target of Mark Raison



Choosing idea: Target of Mark Raison

- Red idea: main ingredient of an innovative project
  - Innovative and easy to develop
- Blue idea: enrich the red idea and reinsure stakeholders
  - Established idea and easy to develop
- Yellow idea: Give a challenge and an extra innovative layer
  - Innovative but challenging to develop

TECHNICAL SPECIFICATIONS

Some technical specifications

- Google Colab



- Jupyter notebook (anaconda)



- Python Command line



Some technical specifications

PITCHING

Pitch

- Pitch that you will give to anyone who asks you who you are and what you do.
- Conferences, seminars, coffee, appointments, telephone prospecting,, ...
- Clear, concise and interesting
- Just flip the question back and start a conversation.

Template

Framework de Geoffrey Moore *(Crossing the Chasm, 1991)*

For *(target customer)*

Who *(statement of need or opportunity)*

The *(product name)* **is a** *(product category)*

That *(key benefit, reason to buy)*

Unlike *(primary competitive alternative)*

Our product *(statement of primary differentiation)*

Template

Exercise!

- Example (coming from a winning idea from Hackathon)

	Catégories à remplir
Pour	Les agriculteurs en herbe
Qui souhaitent	Cultiver des legumes en ville
L'application	Potager partagé
Est un(e)	Application d'allocation de parcelles
Qui offre	À ses utilisateurs un moyen simple de trouver un endroit pour faire pousser ses legumes en ville
Par rapport à	Une recherche traditionelle de potager
Nous	Offrons une interface simple d'utilisation qui permet de trouver une parcelle rapidement et facilement

A few tricks …

- Generate interest right from the start
  - Asking a question
  - Bouncing back on the news
  - Telling a story that speaks to your audience
- Find a persona
- One idea = One slide
- Think about who your audience is and get their attention!



Intermediary Pitch

- **Prerequisite**: upload the completed BMC
  - Can be upload in a visual manner or via a word file with bullet points
  - Describe how you proceeded to fill them in (brainstorming, start-point, etc.)

- Make an **intermediary pitch** (max 5 minutes) about your data analytics project, using the Canvas as structuring tool
  - Idea aspects
    - Idea generation and selection
    - Needs from the mobile app
    - Users targeted
  - Technical aspects
    - Data Sources selection
    - Foreseen data analytics used (ML, GM, IV,…)
  - Implementation aspects
    - Planning (What can we expect from your end-product?)
    - Envisioned issues (legal, ethical)

Thank you !

Questions ? abdelkader.ouared@univ-tiaret.dz



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